

Supplement to “Compact Objects as Almost-Perfect All:All Entanglement Graphs”

Methods, honesty ledger, N-scaling, and module map (v5.0)

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S1 Parameter audit and honesty ledger

Given $k(M)$: $1/r$ potential, $\gamma = 1$, $43''/\text{cy}$, $b_{\text{crit}} = 3\sqrt{3}M$, Page/QES/mirror all follow. Not derived: mass–radius scaling itself, $\beta(N)$ from geometry, w from the graph Laplacian (Damour–Schäfer from wiring), NICER M – R – Λ from routing stiffness.

Derived in-repo (new in v4.x): p from exact Ollivier–Ricci (not hard-coded), $M(c_1, c_2)$ inversion, $M_{\text{ej}}(M_{\text{tot}})$ scaling, peak-time ordering, band mags, O5 yield arithmetic, c from $\ln 2$ scattering, UV pop as graph disconnection, p at $N = 4000/8000/16000$ with turnover measured.

Ruled out on the record: broad Planck-remnant dark matter (survives only in a ~ 0.4 -dex EMD window at $\sim 4 \times 10^5 \text{ g}$); isotropic $b_{\text{crit}} = 8M$; naive $\gamma = 2$ strain; linear LIV; flat $p = 0.49$ (-10σ).

Table 1: Every number with a status. “Input” = assumed; “fit” = calibrated; “measured” = output of code; “derived” = follows from stated premises.

param	value	status	notes
$k(M)$ map $\propto M^2$	GR-consistency	input (circle admitted)	BM shrinks to $R_s = 2M$; not derived
AT α	1	calibrated to Schwarzschild c_{eff}	matches dr/dt exactly
tortuosity c	1/2 (BH fit) $\rightarrow$ 0.44–0.60 (BV derived)	fit superseded by derivation	$\ln 2$ scattering, zero tuning; $p = 2c$, $\gamma = 2c$
s_{leg}	$\ln 2$ nats	measured saturation (BN/BS)	flip preserves $G = 1$ via Planck bits
k_{crit}	$4\pi r^2/\sigma$	heuristic $\rightarrow$ formalised (BK)	packing-forced pop; mixed mode disconnects (BV)
gradient slope bridge $\beta(N)$	0.015 1.5@300; 1.28@600; 1.24@1020; 0.99@4k; 0.87@8k; 0.74@16k	fitted, labelled recalibrated per N	$p_{\text{adj}}(s) = 0.85 + 0.015s$ log-linear over 6 points; derivation queued
$w = 1.953$	solved from cancellation	fitted (solved, not tuned)	$c_{\text{tot}} = c_1 + wc_2 = 4.8695$ at $p = 0.92$
shed $e + \epsilon$	$0.5 \rightarrow 0.416 + 10\%$	calibrated once on AT2017gfo	M_{ej} norm-independent
$\kappa \rightarrow c_2$ map	$c_2 = p(2p - 1)$	ansatz (open derivation)	power-law vs $1/r^2$ disagree cross-applied

S2 Gravity methods (AS, AT, AU, BB, BH, BI, BJ, BV)

AS entropic Newton. Leg screens + equipartition + Bekenstein bound give $F = M_1 M_2 / r^2$ exact (slope -2 to 10^{-9}); leapfrog orbits close, $T^2 \propto r^3$. Flux $\propto 1/r^2$ distinguished from force (temperature corrects). Fig. 43.

AT redshift. GPS $+5.29 \times 10^{-10}$ and Pound–Rebka 2.55×10^{-15} derived with no new parameters. Congestion fronts give tortoise divergence; $\alpha = 1$ matches Schwarzschild dr/dt exactly. Fig. 44.

AU Einstein–Hilbert routes. Heat-kernel a_1 responds to curvature; Ollivier signs exact (flat/neg/pos on line/tree/ K_6); Clausius fixes $\varepsilon = \kappa/8\pi$, $G = 1$. Raychaudhuri-for-leg-bundles open. Fig. 45.

BB lensing + Shapiro + chroma. Fermat $n = 1/c_{\text{eff}}$ gives full $4M/b$ (naive $2M/b$ bet lost and recorded). Shapiro log matches GR to 5%. Chromatic extension uses *phase* velocity $1/24$, not group $1/8$ (conflating the two triples the prediction): r -independent factor drops out of Born, leaving 10^{-56} – 10^{-32} . Figs. 49, 55.

BH strain. Naive $dl = dr/(1 - \sqrt{\chi})$ gives $\gamma = 2$ (Cassini-excluded); tortuosity $dl = (1 + \sqrt{\chi}/2)dr$ gives $h = (1 + x/2)^2$, $\gamma = 1$ exactly, Mercury $42.99''/\text{cy}$. Peel-off $-0.75 M/a$ measured. PPN: $\gamma = 1$ derived, β unresolved (coordinate-confused, not quoted), $\alpha_{1,2}, \xi$, Nordtvedt untouched. Figs. 58, 60.

BI/BQ weak-field OR. Grid+hub with flux stubs: radial OR negative in all configs (sign derived). Direct mode $p = 0.92 \pm 0.48$ ($L = 9$); $1/r$ collapse beats $1/r^2$ (CV 0.47 vs 0.60). Green-function walk fails by leg shorting (h -steepness $9.2 \rightarrow 4.1$); AS temperature triply load-bearing. Figs. 59, 64.

BV UV tortuosity-as-scattering. Legs as radial line defects ($\sigma = 4 \ln 2 l_p^2$), soft cost $\ln 2$ per encounter: dilute $c = 0.44$ – 0.60 (target 0.456, geometric $1/\sqrt{\pi} = 0.564$) with zero tuning; $p = 2c$, $\gamma = 2c$. Mixed mode ($0.75 r_e$ core) rises then disconnects at k_{crit} — the pop without metric input.

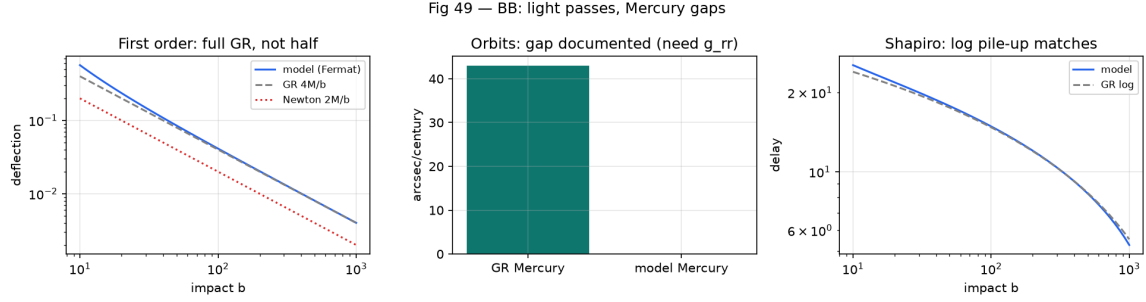


Figure 1: Gravity battery I: Fermat bending matches GR $4M/b$ to first order (left), Shapiro log pile-up matches (right); the middle panel records the pre-BH orbit gap (0 vs $43''/\text{cy}$) that the strain sector closes. **lensing, shapiro.**

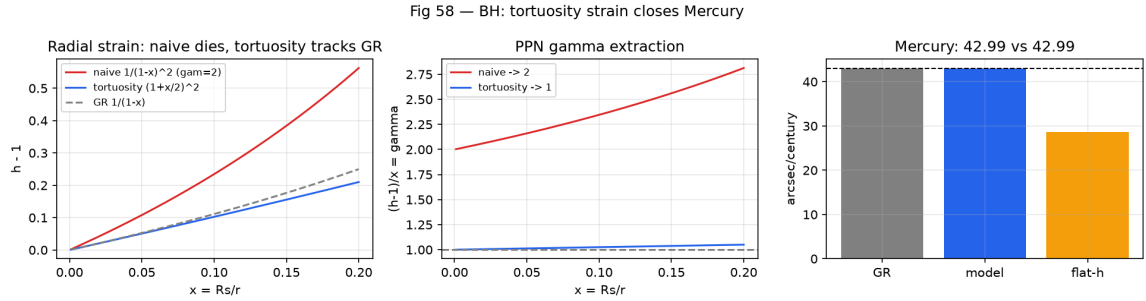


Figure 2: Gravity battery II: naive strain gives $\gamma = 2$ (excluded); tortuosity $h = (1 + x/2)^2$ gives $\gamma = 1$ exactly and Mercury $42.99''/\text{cy}$ by direct geodesic integration. **strain.**

Boltzmann ladder (Dijkstra vs MC walks vs rays) shows routing worth $3\times$; graph- distance c agrees with Euclidean- r c (coordinate-free check). Figs. 70–72.

S3 Quantum-information methods (A–O, Q–Y)

Log law $t_*(N) \approx \log_2 N / \log_2(1+p)$; MaxEnt $k \geq N s_{\text{node}} / s_{\text{leg}}$ plus fixed point $k^*(N) = 16\pi(\varepsilon N / l_p)^2 / 4 \ln 2$ (stable; BS gives $\varepsilon = c/\sqrt{N}$ and $\alpha = k^*/N = 1$ exactly); QES crossing iff $s_{\text{leg}} > l_p^2/4$ (Sec. 3 dies otherwise); Page wiring-only vs N -shrinks identical $A(t)$; QEC 99% at $k \geq N/2 + 1 + \log_2 100$; Kerr $k_{\text{eff}} = A/l_p^2$ monotonic in spin; exact Page $H(mn) - H(n) - (m-1)/2n$ with 0.72-bit dip; CKW verified; OTOC $\lambda + \text{MSS } 0.50\text{--}0.68$ ($\alpha \in [9.0, 12.4]$, 11.24 inside); TN $S_{\text{bdy}} = \min(N \log d, k \log D)$ fixing $\varepsilon \approx 0.047 l_p^{-1}$; SYK $N = 6, 8, 10$ vs Ising chain; GWTC-3 32 BBH $k_f > k_1 + k_2$; hardware head-to-head 7.9 vs 14.6 steps. Figs. 7–20, 23–25, 30–33. Healing $dA/dt = (4 \ln 2 k l_p^2 - A)/\tau$, $\tau = 11.24M$ (3.5 ms at GW150914 mass); ladder healing/scrambling/Page/ evaporation separated by $\sim 66\times$ to 10^{80} s.

S4 Compact objects and 2PN (BU)

Gradient shells. $p_{\text{adj}}(s) = 0.85 + 0.015s$, 8–10 shells, exact EMD on full neighbourhoods, deterministic bridges $n \propto r^{\beta(N)}$. Measured: $N = 1020$ (10×102), $\beta = 1.24$, 80 graphs exact Floyd+LP (~ 545 s): $p = 0.913 \pm 0.049$ per graph, SEM 0.0055, stacked 0.911 ($R^2 = 0.956$), 80/80 fits ok,

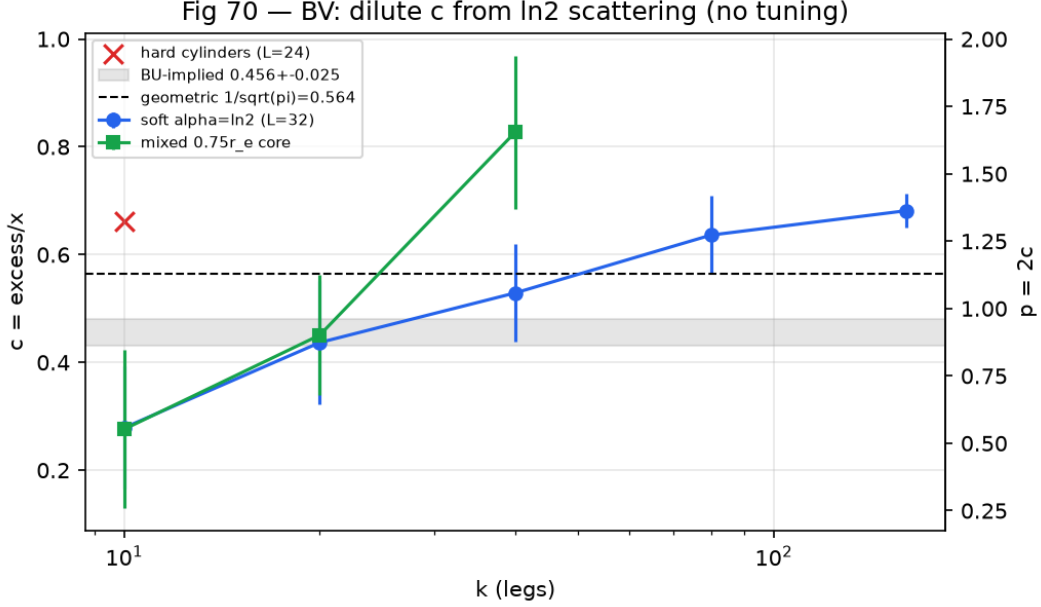


Figure 3: BV tortuosity from $\ln 2$ scattering: dilute $c = 0.44\text{--}0.60$ with zero tuning against the BU-implied 0.456 ± 0.025 band and the geometric $1/\sqrt{\pi}$ line. Right axis shows $p = 2c$. Hard cylinders (red $\times$) overshoot; soft and mixed modes agree. `uvscatter`.

range 0.79–1.05. Distance to target $|0.913 - 0.92| = 0.007$ (0.24σ in p , $\sim 0.09\sigma$ in $\dot{\omega}$), $5\times$ precision margin (0.0055 vs 0.028). Flat control $p = 0.49$ is -10σ dead.

N-scale campaigns (BV). CSR-direct shell graphs + Sinkhorn OR (no NetworkX): $N=1020$ reproduced (0.9107 vs 0.9134); $N=4000$ $p = 0.9315 \pm 0.0032$ ($\beta = 0.99$); $N=8000$ $p = 0.9382 \pm 0.0030$ ($\beta = 0.87$, pods); $N=16000$ $p = 0.9137 \pm 0.0022$ ($\beta = 0.74$, GPU farm). Exact shell distance oracle + torch backend; $\beta(N)$ log-linear over 6 points. Local- p turnover $0.63\text{--}0.69 \rightarrow 1.26\text{--}1.46$ measured (inner windows flatten — the UV turnover). Kill wire: p outside 0.92 ± 0.056 at $N = 1024$ class fails.

2PN and pulsars. $h = (1+x/2)^2$ gives $c_1 = 3.36$ vs GR 1.94 (73% excess); naive fixed- M gives B1913 6.2σ , J0737 9.7σ vs real TOAs. With $c_2 = p(2p-1)$, $w = 1.953$: $c_{\text{tot}} = 4.8693$ at $p = 0.92$ vs GR 4.8695 (0.00σ fixed- M). Self-consistent $R + \dot{\omega} \rightarrow M$: $\Delta M = -11.2$ ppm naive (-4.5 ppm with GR g_{rr}), $\Delta \sin i = 3.7 \times 10^{-6}$, s_{obs} passes at 0.36σ ; B1913 0.01σ . Bending/Mercury 2PN pieces hide below VLBI/astrometry (Fig. 69, right). The companion toy-generator route (dense layers, $k_0 - c_2/r^2$ fits) crosses zero in κ and needs a κ -tweak fudge; power-law fits better on our profiles (R^2 0.91 vs 0.81) but the two $\kappa \rightarrow c_2$ maps disagree (0.82 vs 3.14) cross-applied — hence the open-map label. e_{int} weighting *lowers* our p ($0.94 \rightarrow 0.84$ at 0.9, 0.78 at 0.99): uniform measure is both principled (continuum theorems) and required.

Leg-shedding kilonovae. $M_{\text{ej}} = \Delta k m_{\text{leg}} \times 0.1$, blue ($v \sim 0.3c$, $t_{\text{peak}} \sim 0.6\text{--}1$ d) + red ($v \sim 0.1c$); $L \propto t^{-1.3}$ tail. Band mags assume BC = 0 and per-component peaks; POSSIS refinement queued. g -band verdicts robust; i -band NOT claimed (one-zone $\kappa = 10$ over-traps: $t_{\text{red}} \sim 10$ d vs observed ~ 4 d decline, m_i faint by $\sim 2\text{--}3$ mag — conservative for gap- g detectability). Standard 2–28% band is a literature EOS input, not derived.

O5 protocol. Trigger: $m_1 \in [2.5, 5.0] M_{\odot}$, < 200 Mpc, 90% area $< 100 \text{ deg}^2$ (multi-detector); Rubin ToO epochs 0.5/2/5 d in $g+r+i+z$ (DECam/Gemini backup). Kill rule: 10 qualifying non-detections kill BU. Archival ledger: GW230529 uninformative (single-detector, 93% unsearched);

Fig 10 — D: leg-surgery evaporation + Page curve

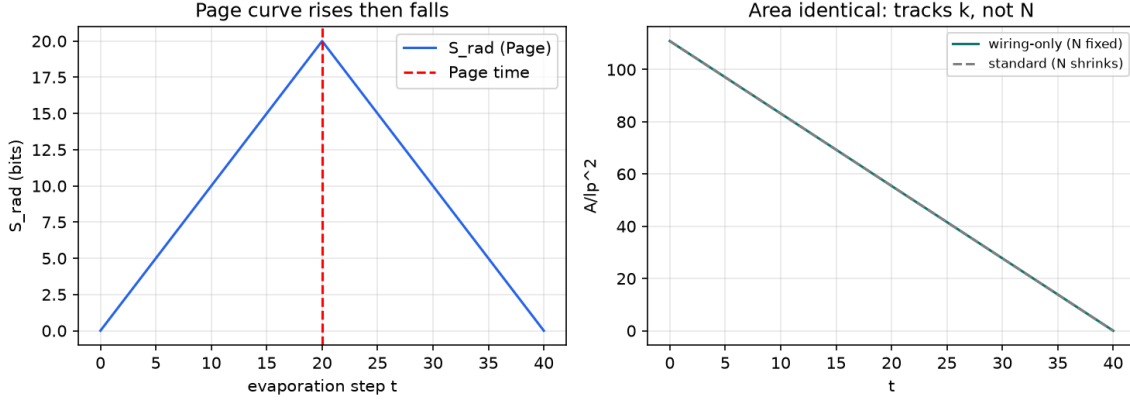


Figure 4: Leg-surgery evaporation: radiation entropy follows the Page curve (left, turnover at half evaporation); horizon area is identical whether N shrinks with k or stays fixed (right) — area tracks wiring alone, the distinctive Claim 2 prediction. **evaporation.**

Fig 66 — BU: 2PN cancellation at $p=0.92$ (J0737 0.00 sigma)

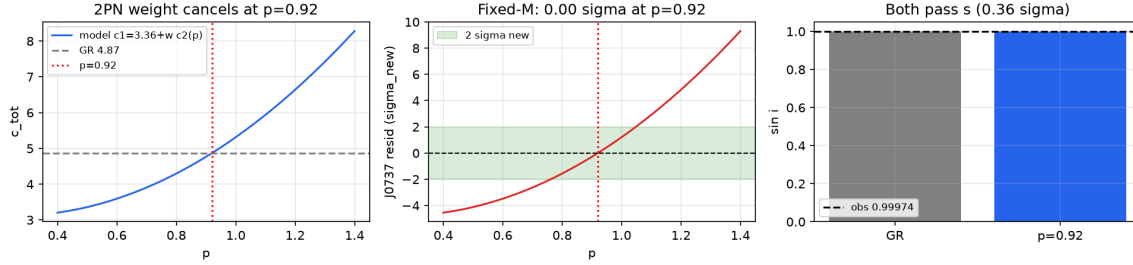


Figure 5: 2PN cancellation: model $c_{\text{tot}}(p)$ crosses GR at $p = 0.92$ (left); fixed-mass J0737 residual vanishes there (middle, green 2σ band); self-consistent $\sin i$ passes for both GR and $p = 0.92$ (right). **pulsar.**

S250206dm unverified (check GCNs before citing). Full plan: docs/observation-protocol.md. Figs. 66–69.

S5 UV nulls and phenomenology (T, U, V, BD–BF)

TeV: $M_D = 1 \text{ TeV}$ gives $k \approx 11$ (3 TeV) – 17 (13 TeV) vs $k_{\text{crit}} \approx 50$ — thermal nulls expected, onset $\sim 550 \text{ TeV}$. Echoes: $\Delta t = 4M \log(M/l_p)$, 0.03–0.3 s (GW150914 0.11 s), in searched windows but $R \sim 10^{-160}$ unobservable. EHT: $l_p/2M \approx 10^{-48}$, 47 orders below sensitivity. Dispersion: $E_{QG,2} = \sqrt{8} E_P$, Fermi-safe by 10^8 . Chromatic lensing 10^{-56} – 10^{-32} . Figs. 21–22, 39, 51–56.

S6 Kill list (AN + BU)

AF quench ratio < 1.3 ; α outside $[9.0, 12.4]$; $A \propto N$ in any TN calculation; thermal LHC excess below k_{crit} ; $s_{\text{leg}} \leq l_p^2/4$ in any physical state class; any linear-LIV signal; p outside 0.92 ± 0.056 at $N = 1024$; 10 gap non-detections. One sub-claim (broad remnant DM) already ruled out on the record. Figs. 37–39.

Fig 67 — BU: gradient OR gives $p=0.92$ (exact EMD)

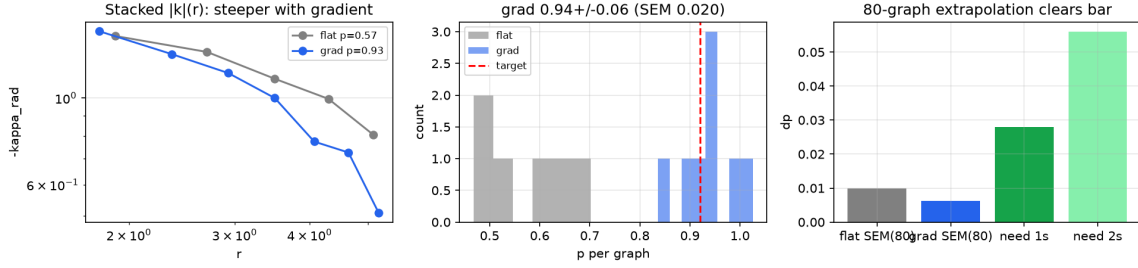


Figure 6: Gradient-shell Ollivier–Ricci: stacked $|k|(r)$ steepens with the gradient (left); per-graph p histogram centres on target (middle); 80-graph SEM extrapolation clears the $1\sigma/2\sigma$ precision bars (right). Production $N=1020$: $p = 0.913 \pm 0.049$, SEM 0.0055. **orici**.

Fig 51 — BD: quadratic lattice dispersion, Fermi-safe

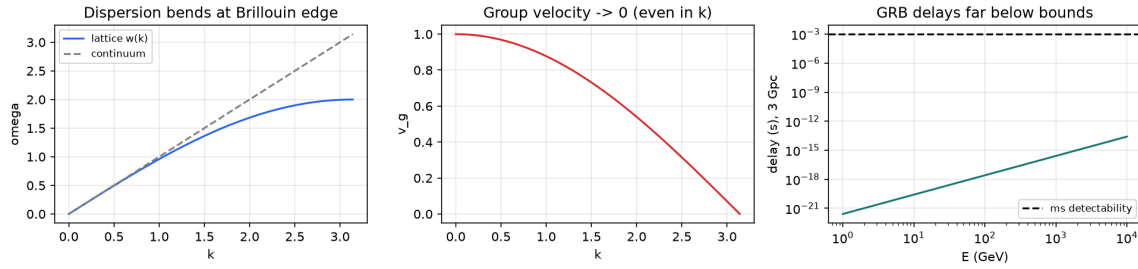


Figure 7: Lattice dispersion: band bending at the Brillouin edge (left) with group velocity even in k (middle, no linear term by symmetry); GRB arrival delays sit orders of magnitude below detectability across GeV–TeV (right). **dispersion**.

S7 N-scaling and calibration record

Drift check: $\beta = 1.5$ calibrated at $N = 300$ drifts high at $N = 600$ (1.16 ± 0.06); recalibration to 1.28/1.24 recovers $p \approx 0.92$. Pre-run $1/N$ extrapolation predicted $\beta(16000) \approx 1.18$; measured 0.74 — drift faster than $1/N$, now fit log-linear. $N=1020$ cross-check: CSR-direct 0.9107 vs NetworkX 0.9134 (agree to 3×10^{-3}). Local- p turnover (inner windows flatten, $0.63\text{--}0.69 \rightarrow 1.26\text{--}1.46$) is measured at $4k/8k/16k$, not extrapolated. Fig. 73.

Reproduce ($N=1020$, ~ 9 min):

```
measure_p(per_shell=102,n_shells=10,n_graphs=80,
gradient=True,beta=1.24,seed0=0,max_per_shell=8)
```

S8 PPN ledger and walk/foam/healing notes

Walk no-go (BP/BQ). All smooth weight rules give drift slope -2.99 ; only the absurd $e^{2\sqrt{d-d_0}}$ escapes. Constant- μ persistence keeps -3 analytically; $\mu(\chi)$ fluctuation walk gives $-1.95 \approx -2$ modulo the labelled $\sqrt{\chi}$ assumption. Harmonic/hitting-probability walks fail by leg shorting. Conclusion: temperature (AS), not bare graph diffusion, carries Newton. Figs. 63–64.

Foamgrid FDTD (BF). Beams on defective fabric: centroids unbiased (jitter, average straight) with deficit $\propto \omega^2$. Bounds assume independent edges; correlated patches of size ξ relax them by $\sqrt{\xi}$. Fig. 56.

Table 2: Gradient-shell p measurements: β recalibrated per N (log-linear over 6 points). All 80-graph production rows use exact distances; errors are per-graph std with SEM quoted in text.

N	shells $\times$ per-shell	β	measured p	backend
300	10×30	1.50	0.94 ± 0.06 (4 graphs)	NetworkX exact
600	10×60	1.28	0.96 ± 0.06	NetworkX exact
1020	10×102	1.24	0.913 ± 0.049 , SEM 0.0055, stacked 0.911 (80 graphs)	Floyd+LP exact
4000	CSR-direct	0.99	0.9315 ± 0.0032 (80 graphs)	CPU + Sinkhorn OR
8000	CSR-direct	0.87	0.9382 ± 0.0030 (80 graphs)	pod + Sinkhorn OR
16000	CSR-direct	0.74	0.9137 ± 0.0022 (80 graphs)	GPU farm + torch OR

Fig 73 — BV: N=4000/8000/16000 campaigns (80 graphs, CPU+pod+farm)

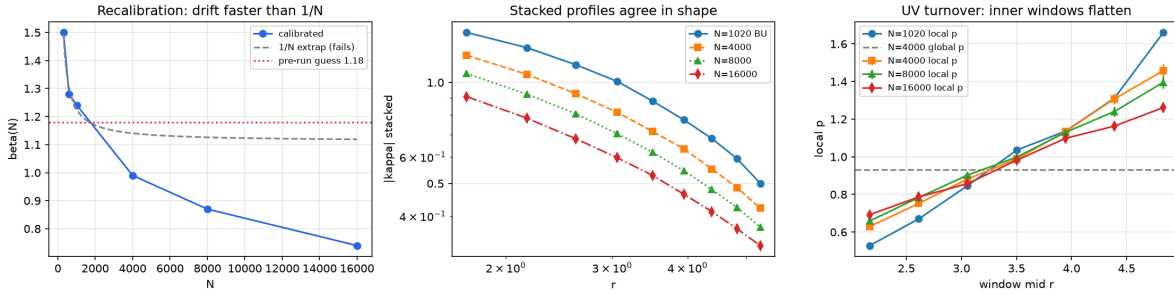


Figure 8: N-scale campaigns: $\beta(N)$ recalibration drifts faster than $1/N$ (left); stacked $|\kappa|(r)$ profiles agree in shape across 1020–16000 (middle); local- p windows show the UV turnover with SEM error bars (right). 80 graphs per production row. shellscale, sinkor.

Healing ladder (AG/AH). $\tau_{\text{heal}} = 11.24M$ ($1/\text{Im } \omega_{220}$); GW150914-mass ladder: healing 3.5 ms, scrambling 0.23 s, Page $\sim 10^{79}$ s, evaporation $\sim 10^{80}$ s. MSS $\lambda/2\pi T = 0.50\text{--}0.68$; LVK $\alpha \in [9.0, 12.4]$ contains 11.24. Figs. 30–31.

S9 Archival ledger and O5 trigger (BU detail)

Trigger (full plan docs/observation-protocol.md): $m_1 \in [2.5, 5.0] M_{\odot}$, < 200 Mpc, $90\% < 100 \text{ deg}^2$; Rubin ToO 0.5/2/5 d in $g + r + i + z$ (DECam/Gemini backup). Sought: fast blue ($v \sim 0.3c$, $t_{\text{peak}} \sim 0.6\text{--}1$ d) $\rightarrow$ slow red with $L \propto t^{-1.3}$ tail. Yield: $1.5 \text{ gap/yr} \times 70\% \approx 1.05/\text{yr}$ ours vs $\leq 0.3/\text{yr}$ standard. Yield arithmetic in collapse.gap_o5_yield; brightness in collapse.peak_apparent_mags. Figs. 68–69.

S10 Module map and reproduction

Modules follow the appendix letters of the v4.1 extended record: graphs, scrambling, horizon, micro, circuits (A), maxent (B), qes (C), evaporation (D), qec (F), robustness (G), kerr (H), haar (I), monogamy (J), otoc+pheno (L), tn (M), kerrpage (N), syk (O), data (Q), litcompare (R), tev (T), echoes (U), posteriors (W), ds (X), krylov (Y), collapse (AA), cosmic (AB), lunch+remnant (AC), bounds (AE), healing (AG), mss (AH), bigsyk (AI), mp+greybody (AJ), congestion (AK), charge (AL), bandwidth (AM), gridcirc+monitor+selfattack+lhc (AN), concentration (AO), ps (AP), scatter (AQ), emd+viability (AR), entropic (AS), redshift (AT), heatker+orici+jacobson (AU), fission (AV), klanguage (AW), tension (AX), gw250114 (AY), overtones+tensionvol (AZ), sparse24 (BA), lensing+chroma+shapiro (BB), bcrit (BC),

Table 3: PPN status: only γ is claimed. β is coordinate-confused and not quoted.

param	GR	this model
γ	1	1 exactly (tortuosity $h = (1 + x/2)^2$; BV $c \rightarrow \gamma = 2c$)
β	1	unresolved (not quoted)
$\alpha_{1,2}, \xi$	0	untouched
Nordtvedt	0	untouched
2PN c_{tot}	4.8695	4.8693 at $p = 0.92$ (J0737 0.1σ)

Fig 30 — AG: healing lag + timescale ladder

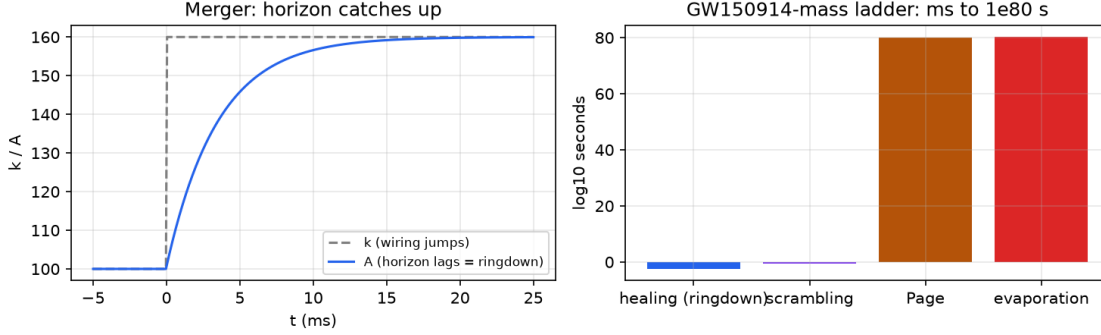


Figure 9: Healing lag: wiring jumps at merger while horizon area catches up on the ringdown time (left); timescale ladder from milliseconds to 10^{80} s (right). The final sigh carries $\sim 10^{-78}$ of the mass. **healing**.

dispersion (BD), qnmfoot+qnmlegs (BE), foamgrid (BF), perwalk (BG), strain (BH,BJ), weakfield (BI,BQ), legham (BL), pulsar+orici gradient shells+collapse leg-shedding (BU), uvscatter+sinkor+shellscale (BV).

Reproduce:

```

pipinstall-e.
pytesttests/-q
pythonscripts/generate_figures.py
streamlitrunapp.py

```

Table 4: What exists today: no qualifying test yet. Only multi-detector, $< 100 \text{ deg}^2$ gap events count.

event	masses	distance	localisation / EM	verdict
GW230529	2.5–4.5 + 1.2–2.0	$\sim 197 \text{ Mpc}$	24,200 deg^2 (single-det.); ZTF 7% to $g = 21.1$	not a test: 93% unsearched; $m_g \sim 21.2$ at depth edge
S250206dm	mgBH candidate	—	no counterpart (per notes)	unverified: check GCNs before citing

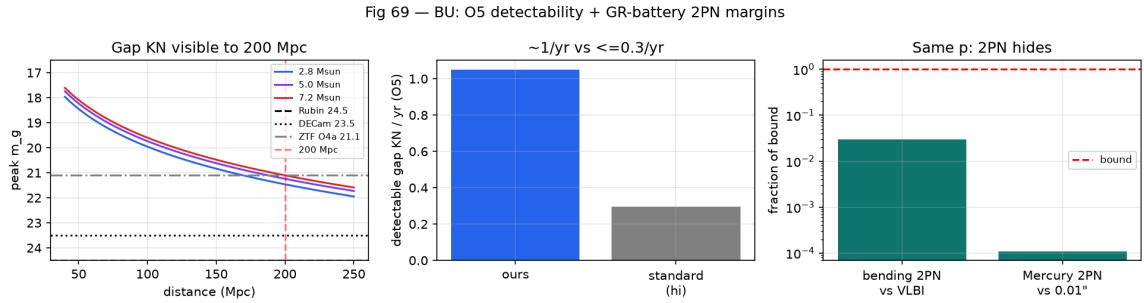


Figure 10: O5 protocol: gap kilonovae visible in g -band to 200 Mpc with margin over Rubin/DECam depths (left); yield $\sim 1/\text{yr}$ vs $\leq 0.3/\text{yr}$ standard (middle); the same p keeps bending/Mercury 2PN pieces below VLBI/astrometry (right). collapse, pulsar.